

COARSE WAGE SETTING AND BEHAVIORAL FIRMS

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Abstract—This paper shows that the bunching of wages at round numbers is partly driven by firm coarse wage setting. Using data from more than 200 million new hires in Brazil, I first establish that contracted salaries tend to cluster at round numbers. Then, I show that firms that tend to hire workers at round-numbered salaries have worse market outcomes. Next, I develop a wage-posting model in which optimization costs lead to the adoption of coarse rounded wages and provide evidence supporting two model predictions using two research designs. Finally, I examine some consequences of coarse wage setting for relevant economic outcomes.

I. Introduction

THE wage-formation process is a central element of many economic models. Although standard wage-formation models assume that workers and firms behave optimally, recent findings challenge this assumption. Both survey and administrative data reveal a tendency for wages to cluster—or “bunch”—at round numbers (Riddles et al., 2016; Dube et al., 2020). This puzzling finding suggests nonstandard behavior by some market participants. According to one view, the bunching is driven by strategic behavior on the firm side, whereby firms pay round-numbered wages to exploit a worker behavioral bias (e.g., left-digit bias). Alternatively, the bunching might reflect the behavior of firms engaging in nonstandard wage setting, possibly due to misoptimization.¹

In this paper, I use rich worker-firm matched data to assess whether the wage bunching is partly due to firm nonstandard wage setting. First, I establish the existence of substantial bunching at round-numbered salaries in the data. Then I provide a set of reduced-form results compatible with firms engaging in coarse wage setting and inconsistent with firms paying round-numbered wages to exploit a worker bias. Motivated by the reduced-form findings, I develop a wage-posting model in which firms pay coarse rounded salaries

due to optimization costs. The model delivers two predictions for which I find support using two research designs. Finally, I quantify some of the downstream consequences of coarse wage setting for within-firm wage inequality, nominal wage stickiness, and policies that affect the wage distribution, such as changes in the minimum wage.

I use an administrative employee-employer matched data set covering the universe of formal-sector firms in Brazil from 2003 to 2017. This data set contains the salary at which firms hire workers (“contracted salary”). I use data on the contracted monthly salary of more than 200 million new hires. In addition, I use a sample of more than 300,000 firms that includes information on all of their employees. Using these data, I document the existence of substantial bunching at round-numbered salaries (i.e., those divisible by ten) in the distribution of contracted salaries. For example, 29.5% of new hires’ contracted salaries are round numbers (a uniform distribution would imply 10%). My findings stand in opposition to the predictions of canonical wage-determination models, in which market-level wages should be smoothly distributed.

Then, I present a series of reduced-form results to shed light on whether worker or firm nonstandard behavior drives the bunching. I begin by identifying a set of firms (“bunching firms”) that tend to hire workers at round-numbered salaries. Next, I compare the market outcomes of bunching firms with those of nonbunching firms. If bunching firms pay round-numbered wages to exploit a worker bias, one would expect these firms to have better market performance than nonbunching firms. However, I find that conditional on a large set of controls, bunching firms tend to experience worse outcomes. For instance, they have worse worker-firm matches as measured by new hires’ separation likelihood and a lower job growth rate, and they are more likely to exit the market.

These reduced-form results suggest that bunching firms do not pay round salaries to exploit a worker bias. Alternatively, firms may pay round-numbered salaries as a simple but coarse approximation when they are uncertain about what the fully optimal salary is. When hiring a new worker, firms face considerable uncertainty about a worker’s marginal revenue product (or “productivity”). Estimating a worker’s contribution to the firm requires answering complex questions: What are all of the possible tasks that the new hire will perform? How does each of these tasks affect the firm’s bottom line? How likely is the prospective employee to successfully accomplish each of these tasks? Instead of attempting to gather all of the information required to compute worker productivity, firms might rely on a rule-of-thumb or heuristic as an approximation—a form of pricing that I refer to as “coarse wage setting.”

As a suggestive reduced-form test for the use of coarse wage setting, I assess whether bunching firms also rely on

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¹Throughout the paper, I use the term “nonstandard” to refer to any behavior that departs from the predictions of the neoclassical model. Specifically, “nonstandard wage setting” refers to firm wage-setting practices that depart from the first-order condition of canonical wage-formation models. See appendix C for a description of the two main classes of wage-setting models.

coarse figures when deciding on salary increases. This is a different environment in which firms also face uncertainty about the optimal action. The canonical model predicts that workers' wage increase depends on their realized productivity (Jovanovic, 1979). This variable is difficult to measure, and so some firms might use coarse approximations as salary increases, such as integer numbers if the salary increase is measured in percentage terms or round numbers if the increase is measured in monetary units. I find that bunching firms also rely on coarse approximations while deciding wage increases. Bunching firms are 29.3 percentage points more likely to offer a round-numbered salary increase in monetary units (from a baseline of 35.4%) and 16.2 percentage points more likely to offer an integer salary increase in percent terms (from a baseline of 34.4%).

The reduced-form results motivate the hypothesis that coarse wage setting is partly what drives the bunching observed in the data. To further explore this hypothesis, I build a wage-posting model in which coarse wage setting is a consequence of optimization costs. The goal of the model is to account for the bunching observed in the data and to generate ancillary predictions that should hold if firms engage in coarse wage setting. The model relies on two key assumptions that I motivate based on numerical cognition research: first, that firms use a rounding heuristic to form an initial estimate of the fully optimal salary (the salary firms would pay if there were no optimization costs), and second, that firms can generate a more precise estimate of the fully optimal salary by paying some cost. The standard wage-posting model is a special case of the model with frictions, in which the optimization cost is zero.

The model delivers two testable predictions that characterize the conditions under which firms are more likely to hire workers at coarse round-numbered wages. First, a smaller expected gap between the coarse wage and the fully optimal wage should increase the likelihood of firms paying the coarse wage. In the model, the firm's benefit of fully optimizing is proportional to this gap. Hence, as this benefit decreases, firms are less likely to pay the cost of computing the fully optimal wage. Second, firms with lower optimization costs should be less likely to pay coarse wages and more likely to pay the fully optimal wage.

I test the model's predictions using two research designs. The first empirical strategy is a bunching design that uses standard techniques in the bunching literature (Kleven, 2016). This design consists of correlating the fraction of workers hired through a coarse rounding heuristic with firm and worker characteristics. I partition the data based on firm and worker characteristics and recover the fraction of workers hired through a coarse wage setting using the "excess mass" in the density of workers earning round-numbered salaries. Second, I estimate linear probability models with firm fixed effects, where the dependent variable is an indicator for paying a round-numbered salary to a new hire. This design allows me to control for a large set of confounding factors, including unobserved heterogeneity at the firm level.

The two research designs deliver similar results in support of the model's predictions.

Although my findings are consistent with nonstandard firm wage setting, they do not imply that firms are making mistakes. For instance, firms may be adopting management practices that, despite leading to coarse wage setting, may improve overall firm performance. Still, these findings are important for three main reasons. First, round-numbered wages make up a disproportionate share of all wages. Thus, understanding why firms pay round-numbered wages sheds light on overall firm wage setting and can inform the modeling assumptions of wage-setting models. Second, research designs that infer parameter values from firm optimality conditions might yield biased estimates due to coarse wage setting. Researchers within the structural tradition often infer unobservable variables using the firm's first-order conditions (FOCs).² However, if firms do not fully optimize with respect to wages, the FOCs may not characterize firms' pricing decisions. Third, coarse wage setting can have downstream consequences for important economic outcomes. I show that coarse wage setting may lead to within-firm wage compression and may increase wage stickiness. In addition, in the presence of firms that engage in coarse wage setting, policies that affect the earnings distribution can affect firm wage-optimization behavior.

This paper contributes to empirical studies of firm wage setting.³ At least since Jones (1896), labor economists have documented the bunching of salaries at round numbers. Some work has suggested that this bunching arises from measurement error (e.g., Schweitzer & Severance-Lossin, 1996). I contribute by documenting bunching in an administrative data set where earnings are not self-reported, which shows that round-number bunching is a real feature of labor markets.⁴ The most closely related paper is Dube et al. (2020). Using unemployment insurance records from the United States, they document substantial bunching at \$10 per hour and show that this pattern is not explained by worker left-digit bias. This raises the possibility that firm nonstandard behavior drives the bunching. I contribute in two main ways, first by establishing a novel set of stylized facts about firms that frequently hire workers at round-numbered wages, and second by quantifying some downstream consequences for such coarse wage setting on important outcomes.

²For example, in the context of a wage-posting model, a researcher equipped with wage data and an estimate of worker productivity could use a firm's FOC to identify the labor supply elasticity. This particular strategy has gained traction in recent years as researchers are increasingly interested in understanding imperfect competition in the labor market (e.g., Lamadon et al., 2022).

³See, among others, Hall and Krueger (2012); Caldwell and Harmon (2019); Hjort et al. (2026); Cullen et al. (forthcoming); Hazell et al. (2026); Lachowska et al. (2022).

⁴Other papers have also documented the bunching of earnings at round numbers (e.g., Kleven & Waseem, 2013; Devereux et al., 2014; Mavroukantis & Seibold, 2022). In these papers, the excess mass of salaries at round numbers is a nuisance parameter, not the object of interest.

This paper also contributes to an emerging empirical literature on simplified firm pricing strategies. The view that firms set prices based on heuristics and simplified rules dates back to Simon (1962), who noted that “price setting involves an enormous burden of information gathering and computation that precludes the use of any but simple rules of thumb as guiding principles.” Recent empirical work provides evidence to support Simon’s claim. For example, Cho and Rust (2010) show that car companies charge a uniform rental price across cars with heterogeneous odometer values, Cavallo et al. (2014) find that global retailers engage in uniform pricing across heterogeneous countries, and DellaVigna and Gentzkow (2019) show that U.S. retail chains engage in uniform pricing across heterogeneous outlets.⁵ Although these papers focus on the goods market, I contribute by documenting a form of simplified pricing in the labor market.

Finally, this paper contributes to the literature studying market outcomes in the presence of behavioral firms. Compared to the ever-growing number of papers that document biases in individuals’ behavior, work on firm heuristics and biases is scarce.⁶ This is partly due to data limitations. Most of the heuristics and biases body of work studies individuals’ behavior in carefully controlled lab environments. There is not a straightforward way of conducting the same type of experiments using firms as research subjects. I contribute by providing field evidence on firm nonstandard behavior in a high-stakes setting.

II. Institutional Context, Data, and Descriptive Statistics

This section provides institutional context on Brazil’s labor market, describes the administrative data set, and provides descriptive statistics of the samples.

A. Brazil’s Labor Market

Brazil’s labor market has both a formal and an informal sector (see appendix B.1). I focus on the formal sector, which employs about 80% of wage employees and has a strict labor code. The contracts of formal-sector workers are governed by the Brazilian Labor Code, which mandates provisions such as a relatively high minimum wage, an extra monthly salary annually, one month of paid leave each year, and high firing costs.

⁵Other work shows that many firms follow coarse pricing policies (e.g., Matejka, 2016; Stevens, 2020).

⁶See Heidhues and Köszegi (2018) for a theoretical overview of this literature and Kremer et al. (2019) for work on behavioral firms in developing countries. Among the empirical papers that study behavioral firms, previous work has shown that entrepreneurs are overconfident regarding future growth (Landier & Thesmar, 2008), restaurant owners do not account for the transitory nature of weather shocks (Goldfarb & Xiao, 2019), car dealerships exhibit loss aversion (Pierce et al., 2025), and retailers underestimate the degree of consumers’ left-digit bias (Strulov-Shlain, 2023). A closely related body of work documents firms’ failure to maximize profits (e.g., Bloom et al., 2013; Hanna et al., 2014; Almunia et al., 2024).

B. Data: Employee-Employer Matched Information

The main data source is the *Relação Anual de Informações Sociais (RAIS)*, an employee-employer matched data set covering the universe of formal-sector jobs in Brazil from 2003 to 2017. This administrative data set is assembled yearly by the Ministry of Labor with information provided by firms. Accurate reporting in the RAIS is required for workers to receive payments from some government programs, and firms face financial penalties for not reporting.⁷

The RAIS contains both firm- and worker-level information. Firms’ data include the number of employees, industry, and location. Workers’ data include educational attainment, occupation, and employment information, including the hiring date, recruitment type (e.g., new hire, transfer, etc.), and contracted salary.

The contracted salary of a worker is central to the empirical analysis. It is the salary contained in an individual’s “Work and Social Security Card” (or CTPS) at the end of each year.⁸ The CTPS documents a worker’s employment history, including the initial salary at the firm and any subsequent modifications. For a new hire, the contracted salary is the initial salary at which the firm hired the worker. For other workers, the contracted salary might differ from the initial salary due to a raise or promotion, for instance.

C. Samples and Descriptive Statistics

New-hires sample. For much of the empirical analysis, I use a *new-hires sample*, in which each observation represents a new hire (defined by a worker-firm-hiring date triplet). To construct this sample, I include only new workers hired each year by private-sector firms. For workers holding multiple positions at a firm in a given year, I keep only the highest-paying position. I exclude new hires without a valid identification number, as well as those with a reported contracted salary below the federal monthly minimum wage. Finally, I keep only new hires with a monthly earnings contract. This excludes, for example, workers who bill by the hour or per day worked, which constitute a small fraction of workers in the data. After imposing these restrictions, the database contains information on the contracted salary of 206 million hires (henceforth, “contracts” or “workers” for short) from 2003 to 2017.⁹

Firm random sample. To conduct any analysis that requires exploiting the panel structure of the data set, I select a random sample of firms. To construct this sample, I create a

⁷The main drawback of the RAIS is that it only contains information on formal-sector workers and firms. Thus, the analysis is not representative of the informal sector. Appendix B.1 uses data from the Brazilian Household Survey, which includes data on informal-sector workers, to compare workers in the RAIS to workers in the entire labor force.

⁸Appendix D.1 provides an example of a CTPS and the information it contains.

⁹Appendix D.4 provides more detail on each of these steps and shows the fraction of excluded observations after each sample restriction.

TABLE 1.—SUMMARY STATISTICS ON WORKERS IN THE RAIS, NEW-HIRES SAMPLE, AND FIRM RANDOM SAMPLE

	RAIS (1)	New-hires sample (2)	Firm random sample (3)
Panel A: Worker characteristics			
Average age	34.966	30.304	32.845
Male	0.580	0.635	0.619
White	0.606	0.574	0.600
Elementary school or less	0.376	0.410	0.408
Completed high school	0.471	0.532	0.510
Completed university	0.153	0.058	0.082
Panel B: Earnings			
Mean monthly salary (reals)	1,288.915	807.502	1,066.613
Median monthly salary (reals)	742.429	621.422	686.848
Share of mean earnings divisible by 10	0.037	0.072	0.041
Share of mean earnings divisible by 100	0.017	0.034	0.018
Share of mean earnings divisible by 1,000	0.003	0.006	0.003
Share of contracted earnings divisible by 10	0.180	0.295	0.230
Share of contracted earnings divisible by 100	0.066	0.121	0.084
Share of contracted earnings divisible by 1,000	0.010	0.019	0.013
Panel C: Industry			
Primary sector	0.046	0.034	0.026
Construction and utilities	0.140	0.167	0.195
Manufacturing	0.121	0.174	0.153
Retail	0.257	0.359	0.368
Services	0.437	0.265	0.259
Panel D: Region			
North	0.054	0.050	0.048
Northeast	0.176	0.149	0.156
Southeast	0.506	0.521	0.530
South	0.173	0.187	0.189
Midwest	0.092	0.093	0.077
Number of observations	842,196,095	206,685,308	28,087,000

This table shows summary statistics on workers in the *Relação Anual de Informações Sociais* (RAIS), the new-hires sample, and the firm random sample. See section II for sample definitions. Earnings are expressed in Brazilian reals.

registry of all private-sector firms ever observed in the RAIS who hired at least one worker at a monthly earnings contract during 2003–2017. Because of computational constraints, I randomly select 5% of them. I track all of the employees of these firms over time (both new hires and other employees). This sample includes more than 300,000 firms, 1.8 million firm-years, and 31.8 million worker-years, of which 28.1 million are at monthly-earnings contracts, the worker-level sample used in table 1 and throughout the analysis.

Descriptive statistics. Table 1 presents summary statistics of workers in the RAIS and the samples. The average worker in the new-hires sample is 30.3 years old. Most workers are male (63.5%) and white (57.4%) and have completed high school (53.2%). The average monthly salary is R\$807 (R\$ signifies Brazilian reals; approximately \$373). Most workers are employed by the retail industry (35.9%), followed by the services industry (26.5%). Workers in the firm random sample have similar characteristics.

III. Bunching in the Distribution of Contracted Salaries

The two main classes of wage-formation models in labor economics are wage-posting models and wage-bargaining

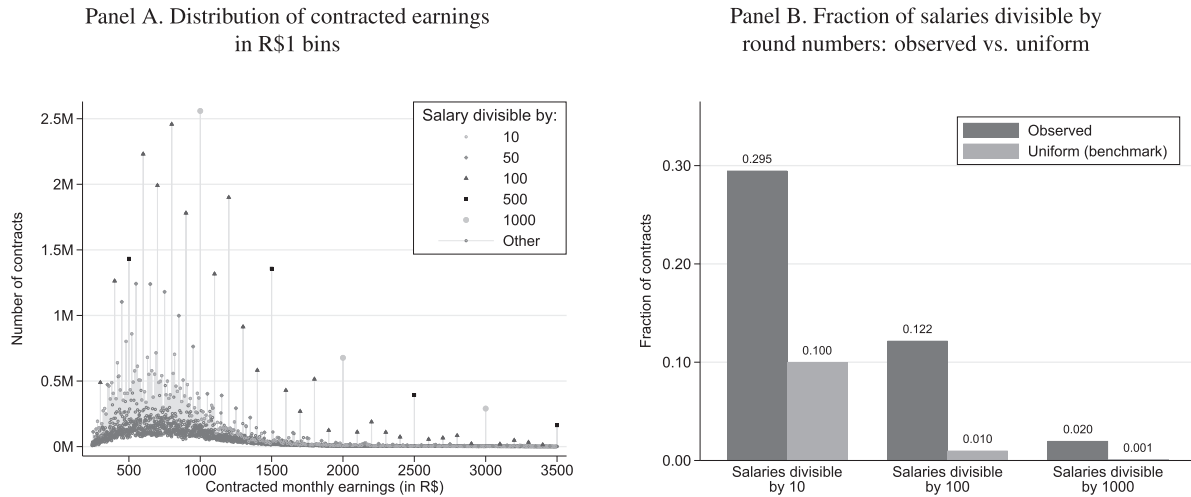
models (Manning, 2011). Under standard assumptions, both types of models predict a smooth distribution of wages at the market level (see appendix C.1).

The data unequivocally reject this prediction. Figure 1, panel A plots the distribution of contracted salaries in the new-hires sample. The earnings distribution exhibits stark bunching at round numbers (i.e., numbers divisible by 10). For example, workers are fifteen times more likely to earn exactly R\$3,000 per month than any other salary between R\$3,001 and R\$3,010. The modal monthly salary in the new-hires sample is R\$1,000, followed by R\$800, and R\$600 (jointly accounting for more than six million contracts), with all being round numbers.

The bunching is also manifested in a nonuniform distribution of the last digit of salaries. Figure 1, panel B shows the fraction of salaries that are divisible by 10, 100, and 1,000. About one-third of the salaries (29.5%) in the new-hires sample are divisible by 10 (see also appendix figure A1). This figure would be 10% if the last digits of salaries were uniformly distributed. More than one-tenth of salaries (12.1%) are divisible by 100 (a uniform distribution would imply 1%), and 1.9% are divisible by 1,000 (a uniform distribution would imply 0.1%).

These figures likely underestimate the true degree of bunching for several reasons. First, the contracted salary

FIGURE 1.—BUNCHING AT ROUND NUMBERS IN THE SALARY DISTRIBUTION



Panel A shows the distribution of contracted salaries in the new-hires sample pooling all of the years during 2003–2017. To construct this figure, I group workers in R\$1 bins and count the number of workers in each bin. Workers whose contracted salary is a round number are denoted with shaped markers. The figure displays only workers with earnings above the minimum wage and below R\$3,500 (which corresponds roughly to the 99th percentile of the distribution of earnings above the minimum wage). Panel B shows the fraction of contracted salaries divisible by 10, 100, and 1,000 in the new-hires sample (dark bars) and the fraction that would be observed if the distribution of the last digits of salaries were uniform (light bars). The figure excludes workers hired at the minimum wage. See appendix D for the sample restrictions.

might be a round number at a different periodicity. For instance, more than one-tenth (10.7%) of the salaries that are not round numbers at the monthly level are round numbers at the yearly level. Similarly, some non-round-numbered salaries might be due to firms setting the wages of new hires equal to the wage of current employees. Some of these wages might have started initially as a round number but were updated over time into non-round-numbered wages.

A. *Bunching of Salaries at Round Numbers in Four Other Data Sets*

To provide further evidence on the existence and magnitude of bunching, I study the earnings distribution in four additional data sets: the 2013 Brazilian Household Survey (Pesquisa Nacional por Amostra de Domicílios, PNAD), the 2013 Brazilian Labor Force Survey (Pesquisa Mensal de Emprego, PME), the 2010 Brazilian Population Census (Censo Demográfico), and the 2013 Social Programs Registry of Individuals (Cadastro Único).¹⁰ In all data sets, I focus on the monthly salary of full-time workers aged 18–65. I exclude workers employed by public-sector firms and individuals who work without remuneration.

The advantage of these data sets is that they include information on informal-sector workers, while the main disadvantage is that earnings are self-reported. Hence,

¹⁰The PNAD is a nationally representative survey conducted annually by the National Statistics Office to measure several characteristics of the population, such as household composition, education, and income. The PME is a monthly survey conducted in six large metropolitan areas to provide frequent updates on the unemployment rate and other labor-market variables. The census is conducted approximately every ten years to count the population in the country, but it also includes earnings information. Finally, the Social Programs Registry contains information on all beneficiaries of government programs, including their earnings.

worker salaries might be measured with error due to—for example—recollection bias or social-desirability bias. Another drawback is that the labor income measure refers to the earnings during the month before the survey was conducted and not the contracted earnings when the employer hired the worker.

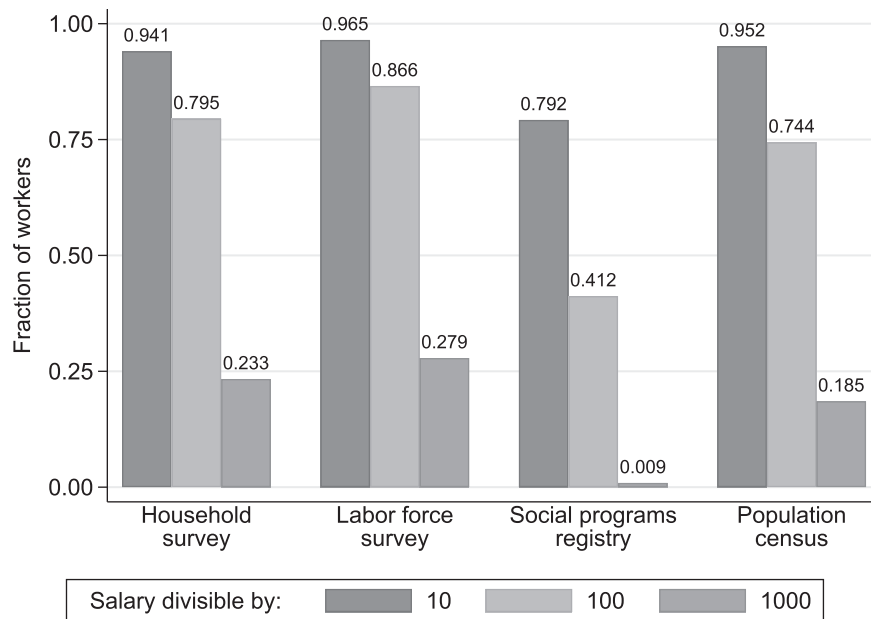
Figure 2 shows the fraction of monthly earnings divisible by 10, 100, and 1,000 in each data set (see appendix figure A2 for the earnings distribution). All data sets exhibit stark bunching at round numbers. For example, 96.1% of monthly earnings in the census are divisible by 10. The corresponding figure in the Household Survey is 94.1%, in the Labor Force Survey it is 96.5%, and in the Social Programs Registry it is 79.2%. This provides additional evidence against the hypothesis that salaries are smoothly distributed. The fact that we do not observe such an extreme bunching in the RAIS is consistent with previous research showing that the bunching in surveys is partly driven by recollection bias from the respondent side, although it could also reflect informal-sector firms paying round-numbered salaries at a higher rate.

Taken together, the results of this section show that bunching at round numbers is a ubiquitous feature of labor markets and not simply a consequence of measurement error.

IV. Firm Nonstandard Behavior and Wage Bunching

This section investigates whether the bunching observed in the data is driven by nonstandard behavior of workers or firms. The approach taken here involves studying the characteristics and outcomes of firms that tend to hire workers at round numbers and assessing whether these are consistent with the hypothesis that the firms that pay round numbers do so to exploit a worker bias.

FIGURE 2.—FRACTION OF SALARIES DIVISIBLE BY ROUND NUMBERS IN FOUR BRAZILIAN DATA SETS



This figure shows the fraction of monthly salaries divisible by 10, 100, and 1,000 observed in four data sets. The data sets are the 2013 Brazilian Household Survey (Pesquisa Nacional por Amostra de Domicílios, abbreviated PNAD), the 2013 Brazilian Labor Force Survey (Pesquisa Mensal de Emprego, abbreviated PME), the 2010 Brazilian Population Census (Censo Demográfico), and the 2013 Social Programs Registry of Individuals (Cadastro Único). The sample comprises full-time workers aged 18–65. I exclude public-sector workers and individuals who work without remuneration.

A. Defining Bunching Firms

I start by measuring a firm's propensity to hire workers at round-numbered salaries. As a simple and intuitive measure, I compute the fraction of a firm's new hires over 2003–2017 whose initial salary is a round number.¹¹ I focus on hiring at salaries divisible by ten for consistency with previous research on the clustering of wages at round numbers (e.g., Riddles et al., 2016) and show that the results below are robust to defining bunching firms using coarser salaries (e.g., those divisible by 100).

Round-number wage setting is highly heterogeneous across firms, with many firms hiring only workers at round-numbered salaries (appendix figure A3). In the data, one in six firms (16.9%) exclusively hired workers at round salaries. I refer to these as *bunching firms*. Appendix table A1 compares the characteristics of bunching and nonbunching firms.

The fraction of bunching firms is higher under less stringent definitions. For instance, 33.2% (27.1%) of firms hired more than half (two-thirds) of their new workers at a round salary. In appendix tables A2–A7, I show that the results below are robust to these alternative definitions of bunching firms. The results are also robust to excluding small firms (i.e., firms that employ fewer than five workers) and using the *yearly* salary of new hires to define bunching firms (instead of the monthly salary).

¹¹I exclude from this computation workers hired at or below the minimum wage and those without a monthly earnings contract.

B. Market Outcomes of Bunching Firms

A potential rationale for why many firms pay round-numbered salaries is to extract surplus from workers who have nonstandard preferences. If this hypothesis holds true, one should see the consequences reflected in better firm outcomes. To evaluate this, I consider four outcomes: worker separation and resignation likelihoods during the hiring year or the following year, which I use as proxies of a poor worker-firm match; the growth rate of the firm's size, as measured by its number of employees; and a binary variable indicating if a firm leaves the market. Although I do not observe firm profit, the firm growth and survival rates are functions of realized profits.

In table 2, I estimate regressions of the form

$$y_{ijt} = \alpha + \beta \text{BunchingFirm}_j + \psi X_{it} + \delta Z_{jt} + \varepsilon_{ijt}, \quad (1)$$

where i denotes workers, j firms, and t years; y_{ijt} is one of the four outcomes; and BunchingFirm_j equals one if firm j hired all new employees at a round salary in the sample.

The regression includes X_{it} , a vector of worker characteristics (age, gender, race, and occupation dummies), and Z_{jt} , a vector of fixed and time-varying firm characteristics that are typically associated with firm sophistication (see appendix D.2 for variable definitions). These characteristics are the presence of a human resources (HR) department, the share of employees with a high-school and college degree, educational attainment of the manager, firm age, mean earnings of the firm employees, firm size, and firm hiring experience. I flexibly control for firm size and hiring experience

TABLE 2.—OUTCOMES OF FIRMS THAT TEND TO HIRE WORKERS AT ROUND NUMBERS

	Dependent variable			
	New hire separated (1)	New hire resigned (2)	Firm job growth rate (3)	Firm left market (4)
Bunching firm	0.041*** (0.003)	0.012*** (0.001)	−0.034*** (0.002)	0.011*** (0.001)
Mean dependent variable	0.353	0.115	0.016	0.101
N	20,281,312	20,281,312	1,381,246	1,752,411

This table displays estimates of β from equation (1). Each column shows the result of a regression using the dependent variable listed in the column header. In column 1, the outcome equals one if a new hire separated from the firm during the year she was hired (year t) or the following year (year $t + 1$), and zero otherwise. Column 2 is defined analogously but using worker resignation likelihood instead of separation likelihood. In column 3, the dependent variable is the percent change in the number of workers employed between consecutive years. In column 4, the outcome is a dummy that equals one if the firm had no workers at the end of the year and zero otherwise. I use the firm random sample to estimate all regressions. In columns 1 and 2, the regressions are estimated at the worker-by-firm-by-year level and using only data from the year in which a worker was hired and the following year. In columns 3 and 4, the regressions are estimated at the firm-by-year level. The regressions control for firm age, share of employees with completed high school, share of employees with completed college, educational attainment of the firm manager, a dummy for having an HR department, the mean earnings of the firm employees, firm size fixed effects, number of hires fixed effects, and industry-by-microregion-by-year fixed effects. The specifications in columns 1 and 2 additionally control for worker gender, race, and occupation. Heteroskedasticity-robust standard errors clustered at the firm level in parentheses. * Significant at 10%, ** significant at 5%, and *** significant at 1%.

by including fixed effects for the number of workers hired and the mean number of workers employed. Z_{jt} also includes industry-by-year-by-microregion fixed effects.¹²

To analyze worker separation likelihood, I estimate the regressions at the worker-by-firm-by-year level. To analyze the firm growth and survival rates, I estimate the regressions at the firm-by-year level (and exclude the worker controls). I cluster the standard errors at the firm level.

Table 2 shows that bunching firms tend to have worse outcomes. Columns 1 and 2 show that new hires in bunching firms are, on average, 4.1 percentage points (an 11.6% increase relative to the sample mean) and 1.2 percentage points (or 10.4%) more likely to separate and resign, respectively, than new hires in nonbunching firms ($p < 0.01$). Column 3 shows that bunching firms have a 3.4 percentage points lower growth rate, on average, than nonbunching firms ($p < 0.01$). Column 4 shows that bunching firms are 1.1 percentage points (or 10.8%) more likely to exit the market ($p < 0.01$).

These results are robust to excluding small firms, varying the set of controls, and using alternative definitions of bunching firms. Appendix table A2 estimates the baseline specification for the subset of firms that employ more than five workers, on average across all years. The same results hold for these large firms. Appendix table A3 estimates all specifications at the worker level and additionally controls for the wage level by including wage fixed effects (in R\$100 bins). The point estimates are quantitatively similar to those of the baseline specification. Finally, appendix table A4 shows that the results are robust to defining bunching firms in several alternative ways.

In appendix B.2, I study additional outcomes. Specifically, I analyze whether the higher separation rates of new workers hired by bunching firms persist over time and whether the higher separation rates are driven by high- or low-skilled

workers, and I assess whether the lower job growth rate of bunching firms is driven by high- or low-skilled workers. I find that the higher separation rates persist up to three years after new hires join bunching firms (appendix table B2). The higher separation and resignation rates of new workers hired by bunching firms are mainly driven by high-skilled workers (appendix table B3). Furthermore, bunching firms experience lower job growth rates for both high-skilled and low-skilled workers, as well as for high- and low-paid employees (appendix table B4).

These results admit several interpretations. First, the worse outcomes experienced by bunching firms may partly be a consequence of paying round-numbered salaries. For example, the higher separation likelihood might be due to a poor worker-firm match caused by paying a nonoptimal wage. In addition, the results can also be explained by bunching firms being less sophisticated in other unobserved dimensions, which in turn might drive their worse outcomes. For example, in addition to having nonstandard pay-setting practices, bunching firms may have less efficient production processes, which may be the cause of the lower survival rates. Finally, the round-number wage setting may be the result of the constrained optimization problem faced by firms. For example, managers might (optimally) prioritize spending resources to improve production efficiency over pay-setting strategies. Regardless of the right explanation, the results are at odds with the hypothesis that sophisticated firms pay round-numbered salaries to exploit a worker bias.

C. Behavior of Bunching Firms in a Different Decision Environment

Another possible explanation for why many firms pay round-numbered salaries is that they are uncertain about what the fully optimal salary is and use round-numbered salaries as a simple but coarse approximation.¹³ For example, firms might be uncertain about worker productivity,

¹²A microregion is a geographical area that groups together economically integrated contiguous municipalities with similar productive structures. There are about 500 microregions in Brazil, each of which can be thought of as a local labor market (see Dix-Carneiro & Kovak, 2017). The boundaries of these microregions are defined by the National Statistics Office of Brazil.

¹³Simplified pricing strategies have been documented in several environments (e.g. Cho & Rust, 2010; DellaVigna & Gentzkow, 2019; Stevens, 2020).

TABLE 3.—USE OF ROUND-NUMBERED SALARIES ACROSS DECISION ENVIRONMENTS

	Dependent variable					
	Salary increase in R\$ is a round number		Salary increase in % is an integer		Either a round number or an integer	
	(1)	(2)	(3)	(4)	(5)	(6)
Bunching firm	0.293*** (0.003)	0.322*** (0.004)	0.162*** (0.003)	0.082*** (0.003)	0.286*** (0.003)	0.331*** (0.004)
Mean dependent variable	0.354	0.123	0.344	0.110	0.421	0.214
<i>N</i>	4,953,782	3,646,479	4,953,782	3,646,479	4,953,782	3,646,479
Excluding zero growth	No	Yes	No	Yes	No	Yes

This table displays estimates of β from equation (1). In columns 1 and 2, the outcome is a dummy that equals one if the change in worker's i 's contracted salary between t and $t + 1$, measured in Brazilian reais, is a round number and zero otherwise. In columns 3 and 4, the outcome is a dummy that equals one if the percent change between t and $t + 1$ of worker i 's contracted salary is an integer and zero otherwise. In columns 5 and 6, the outcome equals one if either the absolute wage change is a round number or the percent change is an integer and zero otherwise. I use the firm random sample to estimate all regressions. The regressions are estimated at the worker-by-firm-by-year level and using only data from the year in which a worker was hired and the following year. Even columns exclude new hires whose salary did not change in nominal terms. The regressions control for worker gender, worker race, worker occupation, firm age, share of employees with completed high school, share of employees with completed college, educational attainment of the firm manager, a dummy for having an HR department, the mean earnings of the firm employees, firm size fixed effects, number of hires fixed effects, and industry-by-microregion-by-year fixed effects. Heteroskedasticity-robust standard errors clustered at the firm level in parentheses. *Significant at 10%, **significant at 5%, and ***significant at 1%.

which is a central determinant of the optimal salary in wage-determination models.

If this hypothesis is true, one would also expect these firms to rely on similar coarse approximations in other environments where they also face uncertainty. To evaluate this, I use salary increases as a different domain to explore the potential use of coarse pay setting. In this domain, firms face uncertainty about employee realized productivity. The canonical Bayesian model of wage formation predicts that a worker's wage increase depends on her realized productivity (Jovanovic, 1979; Terviö, 2009). By contrast, coarse pricing suggests that firms determine raises based on coarse approximations, such as integer numbers if the salary increase is measured in percentage terms or round numbers if the increase is measured in monetary units.

Table 3 shows estimates of equation (1) using two measures of coarse pay setting as outcomes: first, a dummy that equals one if a new hire received a round-numbered salary increase in Brazilian reais (e.g., R\$310 as opposed to R\$314), and, second, a dummy that equals one if a new hire received an integer salary increase in percentage terms (e.g., 3% as opposed to 3.14%).

Firms that tend to hire workers at round salaries also tend to rely on coarse figures when deciding salary increases. Columns 1 and 3 show that bunching firms are 29.3 percentage points more likely to offer a round-numbered salary increase in Brazilian reais (from a baseline of 35.4%, $p < 0.01$) and 16.2 percentage points more likely to offer an integer salary increase in percent terms (from a baseline of 34.4%, $p < 0.01$). Column 5 shows that bunching firms are about 28.6 percentage points more likely to engage in either of the two behaviors (from a baseline of 42.1%, $p < 0.01$).

These results are robust to excluding workers whose salaries remained constant in nominal terms (columns 2, 4, and 6). They are also robust to excluding small firms (appendix table A5), controlling for the wage level (appendix table A6), and defining bunching firms in several alternative ways (appendix table A7).

V. Wage-Setting Model with Optimization Frictions

The evidence in section IV motivates the hypothesis that firm coarse wage setting is partly behind the rounding observed in the data. To further explore this hypothesis, in this section I build a wage-posting model in which coarse wage setting is a consequence of optimization frictions. The goal of the model is to account for the bunching observed in the data and to generate additional testable predictions.

This section first reviews evidence from numerical cognition research to support the model's assumptions. Next, I present a summary of the model and discuss the model's testable predictions. Then, I describe the two research designs that I use to test the predictions and present the results.

A. Stylized Facts from Numerical Cognition Research

Round numbers are ubiquitous in open numerical judgments. For example, in contingent valuation studies, individuals often report round numbers (Whynes et al., 2005). Similarly, in judging the likelihood of future events, subjects often report round-numbered probabilities (Manski & Molinari, 2010). According to numerical cognition research, this phenomenon occurs because the mental computation cost of round numbers is lower, and, thus, round numbers are the first that "come to mind." I use this finding to motivate one of the assumptions of the model, namely, that firms use a round number as an initial estimate of the worker fully optimal salary.

Numerical cognition research also sheds light on how individuals generate more precise estimates. According to prominence theory (Albers & Albers, 1983; Albers, 2001), individuals start from a round number and sequentially refine it by adding and subtracting smaller round numbers until they reach a satisfactory estimate.¹⁴ Following these

¹⁴Consistent with this theory, Converse and Dennis (2018) show that individuals are more likely to use "prominent numbers" (a subset of the round numbers) in numerical judgments when they are induced to quickly

findings, in the model I assume that firms can pay some cost to refine their initial estimate of the optimal salary.¹⁵

B. Summary of the Model

This section presents an abbreviated version of the model, focusing on its key assumptions and predictions. Appendix C.2 provides a complete description of the model.

In the model, monopsonistic firms decide what wage to offer to prospective workers. In the textbook formulation of the wage-posting model, firms know the marginal revenue product (MRP) of hiring an additional worker and offer a wage proportional to it. The difference between the textbook model and the one presented here is that I depart from the assumption that firms observe worker MRP.

The model rests on two key assumptions. First, I assume that firms form an estimate of the fully optimal salary (the salary that firms would pay if they had perfect information) based on a coarse rounding heuristic. This assumption is supported by the research on numerical cognition reviewed above. For simplicity, I model hiring decisions around a single round number. Appendix C.3 considers an extension where firms can approximate the fully optimal salary with different degrees of precision.

The second key assumption is that by paying an “optimization cost,” firms can generate a more precise estimate of the fully optimal salary. This reduced-form cost likely reflects a range of underlying mechanisms, including information-gathering costs, attention costs, and the cost of integrating the data available.¹⁶ I say that a worker is hired through coarse wage setting (or a coarse rounding heuristic) if the firm does not pay the optimization cost and instead hires the worker at the round-numbered salary.

Under these two assumptions, the market-level distribution of wages comes from a mixture of two distributions: one distribution with the same support as the distribution of fully optimal wages and one with support on the set of round numbers. The (endogenous) mixture weight is the fraction of workers hired through coarse wage setting, a variable de-

noted by θ . Hence, the cross-sectional distribution of wages in the model exhibits bunching at round numbers. The standard wage-posting model is a special case of the model with optimization frictions, in which the optimization cost is zero (which implies $\theta = 0$).

C. Model Predictions

Firms pay the fully optimal salary whenever the benefit of doing so exceeds the optimization cost; otherwise, they rely on the coarse rounding heuristic and pay a round-numbered salary. The comparative statics generate the following testable predictions:

Prediction 1. As the value of the expected gap between the coarse wage and the fully optimal wage decreases, firms are more likely to rely on the coarse rounding heuristic and pay a round-numbered salary. This is the case because the profit return to generating a more precise estimate of the fully optimal wage is proportional to the distance between this wage and the coarse wage.

To test prediction 1, I exploit changes in the purchasing power of gaps over time and across regions in the country. As inflation erodes the purchasing power of money, the real monetary cost of mispricing a fixed gap decreases. Intuitively, “getting the wage right” is less profitable in real terms.

Prediction 2. Firms with a higher optimization cost are more likely to rely on coarse wage setting and thus pay round-numbered salaries. Intuitively, as finding the fully optimal salary becomes costlier, firms are more likely to rely on a coarse approximation.

The firm optimization cost is unobservable. I use two proxies of firm optimization cost: firm size and hiring experience. Larger firms and firms with more hiring experience might have a lower optimization cost because they are more likely to have an HR department or structured management practices (Cornwell et al., 2021). Thus, to test prediction 2, I evaluate whether a given firm is less likely to pay a round-numbered salary as it grows larger over time or accumulates more hiring experience.

D. Testing the Model Predictions

I use two research designs to test the model’s predictions: a bunching design and a regression design. First, I describe each design and then present the results.

Bunching design. The first research design consists of estimating θ , the fraction of workers hired through a coarse rounding heuristic, for each value taken by an observable variable, such as firm size or worker educational attainment, and testing whether the sign of the correlations aligns with the model’s predictions.

make a judgment and when they are under a high cognitive load. Relatedly, Giustinelli et al. (2020) show that individuals with high cognitive ability are less likely to give round-numbered responses in expectations surveys, possibly because they have a lower cost of refining their estimates.

¹⁵The notion that it is costly to obtain more precise estimates of a target value also has parallels in mathematics and computer science. For example, improving the precision of a Taylor expansion approximation (i.e., computing more decimals) requires increasing the number of expansion terms, requiring more computational power and memory to store the additional terms.

¹⁶The compensation reports sold by pay-consulting firms such as ADP or PayScale provide a market-based approach to quantifying all these costs. These reports provide advice on how much a firm should pay a prospective employee with given characteristics. Appendix figure A4 shows an example of a compensation report. After gathering information on the prospective employee, such as job title, educational attainment, and years of experience, these firms provide a distribution of suggested compensations. It is noteworthy that the suggested compensations in appendix figure A4 are not round numbers.

By definition, θ can be written as the ratio between B , the number of workers hired through coarse wage setting, and N , the total number of new hires:

$$\theta = \frac{B}{N}. \quad (2)$$

Although B is not observed in the data, it can be estimated by assuming that the excess mass of workers at round numbers in the earnings distribution represents workers hired through coarse wage setting. To compute $\hat{B}$, it is necessary to estimate a counterfactual distribution in which there is no bunching, which I obtain using standard techniques of the bunching literature (Kleven, 2016). Appendix E describes the methodology in detail.

The estimated excess number of workers at round number r , $\hat{B}_r$, equals the difference between the number of workers earning r in the actual and the counterfactual distribution, $\hat{B}_r = C_r - \hat{C}_r$. To estimate B , I integrate the excess mass across all round numbers:

$$\hat{B} = \sum_{r \in R} \hat{B}_r, \quad (3)$$

where $R = \{w \mid w = 10k \text{ for some } k \in \mathbb{Z}\}$ is the set of round-numbered salaries. I estimate θ by replacing B in equation (2) with its empirical counterpart, $\hat{B}$:

$$\hat{\theta} = \frac{\hat{B}}{N} = \frac{1}{N} \sum_{r \in R} \hat{B}_r. \quad (4)$$

To test the predictions of the model, I estimate θ conditioning on the values taken by a given covariate. For example, I calculate the excess number of workers in the distribution of college-educated workers and then compute the ratio between this estimate and the total number of college-educated workers. This ratio represents the fraction of college-educated workers who were hired through coarse wage setting. I repeat this process for workers with only a high-school diploma, etc. More generally, this procedure yields estimates of B and θ for each value taken by a covariate of interest. I use this procedure to examine whether $\hat{\theta}$ is correlated with characteristics of the firm (e.g., size or hiring experience).

To evaluate whether a decrease in the value of the gap increases coarse wage setting (prediction 1), I calculate the correlation between $\hat{\theta}$ (estimated for each metropolitan region-month-year triplet) and the log of the consumer price index (CPI) of the corresponding region-month-year in which the worker was hired.¹⁷ To assess whether a lower optimization cost reduces coarse wage setting (prediction 2), I test for a negative correlation between $\hat{\theta}$, firm size, and firm hiring experience. In the bunching design, these correlations are mainly identified off of cross-sectional variation in firm size and hiring experience. Under the assumption that larger

firms and firms with more hiring experience have lower optimization costs, we should observe a negative correlation between these variables and $\hat{\theta}$.

Regression design. The second research design is a regression design that allows me to control for a large set of variables, including unobserved heterogeneity at the firm level. I estimate linear probability models of the form

$$\begin{aligned} \mathbb{1}\{w_{ijst} \in R\} = & \pi \log \text{CPI}_{st} + \beta_1 \text{FirmSize}_{jt} \\ & + \beta_2 \text{HiringExp}_{jt} + \delta X_{it} \\ & + \gamma_j + \gamma_t + \gamma_s + \varepsilon_{ijst}, \end{aligned} \quad (5)$$

where the dependent variable, $\mathbb{1}\{w_{ijst} \in R\}$, equals one if the contracted salary of new hire i employed by firm j in metropolitan region s during month m in year t is a round number, and zero otherwise, FirmSize_{jt} is the (log) number of employees, and HiringExp_{jt} is the (log) number of employees hired since the firm first appeared in the sample. Equation (5) also includes X_{it} , a vector of worker-level characteristics (gender, education, working experience, and occupation), and region, year, and firm fixed effects. I cluster standard errors at the firm level and normalize all covariates by their standard deviation so that their corresponding coefficients can be interpreted as partial correlations. This normalization makes the results of the regression design comparable to those of the bunching design.

I use the coefficients of equation (5) to test the model's predictions. To assess whether a smaller gap—in real terms—reduces coarse wage setting (prediction 1), I test whether $\hat{\pi} > 0$. Given the region and year fixed effects, identification mainly comes from within-region changes in the price level over time. To assess whether a higher optimization cost increases coarse wage setting (prediction 2), I test whether $\hat{\beta}_1 < 0$ and $\hat{\beta}_2 < 0$. Since equation (5) includes firm fixed effects, these coefficients are identified off of variation in the size and number of workers hired by a given firm over time.

Results. Table 4 shows the results. Columns 1 and 2 present the results of the bunching design and columns 3 and 4 the results of the regression design.

The first row shows the relation between hiring workers at a coarse salary and the log CPI (prediction 1). The bunching design shows a positive and statistically significant relationship between the fraction of workers hired through coarse wage setting and the log CPI ($p < 0.01$). Consistent with this, the regression design shows that—ceteris paribus—an increase in the inflation rate increases the likelihood of a given firm paying a round-numbered salary to new hires ($p < 0.01$).

The second and third rows display the relation between hiring workers at a coarse salary and the two proxies of the optimization cost (prediction 2). Larger firms and firms that have hired more workers have a lower likelihood of relying on coarse salaries ($p < 0.01$). Consistent with this, the

¹⁷Metropolitan region is the most disaggregated geographical level at which CPI data are available. The Brazilian National Statistics Office collects inflation data at the monthly level for eleven metropolitan regions. Each metropolitan region is a collection of several municipalities.

TABLE 4.—TESTING THE PREDICTIONS OF THE MODEL

	Dependent variable			
	Fraction of workers hired through coarse wage setting ($\hat{\theta}$)		Dummy for hiring a worker at a round number ($\mathbb{1}\{w_i \in R\}$)	
	(1)	(2)	(3)	(4)
Consumer price index (logs)	0.108*** (0.020)	−0.004 (0.021)	0.036*** (0.001)	0.029*** (0.001)
Firm size (logs)	−0.972*** (0.034)	−0.879*** (0.086)	−0.012*** (0.001)	−0.032*** (0.001)
Firm hiring experience (logs)	−0.882*** (0.050)	−0.053 (0.140)	−0.025*** (0.001)	−0.002** (0.001)
Fixed firm sample?	No	Yes	No	Yes

This table shows linear correlations between the covariate listed in the row header and the outcome listed in the column header. In columns 1 and 2, the dependent variable is the estimated fraction of workers hired through coarse wage setting, $\hat{\theta}$. Section VD describes how $\hat{\theta}$ is estimated. Each observation denotes the excess mass at each value taken by a covariate. Thus, the sample size varies by covariate. For log CPI, $N = 1,980$. For firm size and hiring experience, $N = 100$. In columns 3 and 4, the dependent variable is a dummy that equals one for workers hired at a round-numbered salary ($\mathbb{1}\{w_i \in R\}$). In addition to the variables listed in the table, the regressions control for worker gender, worker race, worker occupation, worker education, worker potential experience, firm fixed effects, year fixed effects, and metropolitan region fixed effects. I normalize variables by their standard deviation so that the coefficients of the regressions can be interpreted as the linear correlation coefficients. The sample size is $N = 3,988,606$ in column 3 and $N = 1,493,286$ in column 4. Heteroskedasticity-robust standard errors clustered at the firm level in parentheses. *Significant at 10%, **significant at 5%, and ***significant at 1%.

regression design shows that as firms grow larger in size and accumulate more hiring experience, they become less likely to hire workers at round-numbered salaries ($p < 0.01$).

A possible concern is that some of the correlations might be partly driven by the fact that firms that tend to hire workers at round-numbered salaries are more likely to exit the market. This type of selective attrition could explain the negative association between firm size and paying coarse wages. To deal with this, in columns 2 and 4 of table 4, I reestimate all specifications using a fixed sample of firms that I observe in all fifteen years of the data. The coefficients tend to be similar to the baseline results, albeit in some cases the magnitudes are smaller and the estimates more imprecise.

As an additional robustness test, in appendix table A8 I test the model's predictions using alternative measures of the dependent variable. Instead of using all round-numbered salaries, I measure the dependent variable using salaries divisible by 100 (panel A) or 1,000 (panel B). The results are remarkably consistent across specifications. For example, the correlation between the fraction of workers hired through coarse wage setting and firm hiring experience is -0.88 in the baseline specification, compared to -0.87 when using salaries divisible by 100, and -0.64 when using salaries divisible by 1,000.

In summary, I find evidence in support of the two predictions of the model using two different research designs. This is consistent with the hypothesis that the bunching of wages at round numbers observed in the data is partly due to coarse wage setting. Appendix F evaluates whether several alternative explanations are compatible with the bunching observed in the data and the stylized facts documented in this section. The alternative explanations that I discuss are worker left-digit bias, focal points in wage bargaining, collective bargaining agreements, fairness concerns, round wages as a signal of job quality, and changes in marginal tax rates. Although some of these explanations have explanatory power in accounting for some features of the data, I conclude that none of them can provide a cohesive account of the entire pattern of results.

VI. Implications for Other Economic Outcomes

In this section I explore some of the downstream consequences of firm coarse wage setting for important economic outcomes.

A. Within-Firm Wage Inequality

Understanding the drivers of wage inequality is an important research agenda in public and labor economics. Previous research has found that firm wage-setting policies influence wage inequality (Card et al., 2018). One might expect firm coarse wage setting to affect wage dispersion among new hires.¹⁸ To assess this, I estimate equation (1) using as outcomes the Gini coefficient and ratios between the contracted salary at the 90th and 10th percentile, 90th and 50th percentile, and 50th and 10th percentile.¹⁹ Equation (1) includes fixed effects for the number of workers hired, and so the research design compares the within-firm wage inequality of two firms that hired the same number of workers using a different decision rule to determine their initial pay.

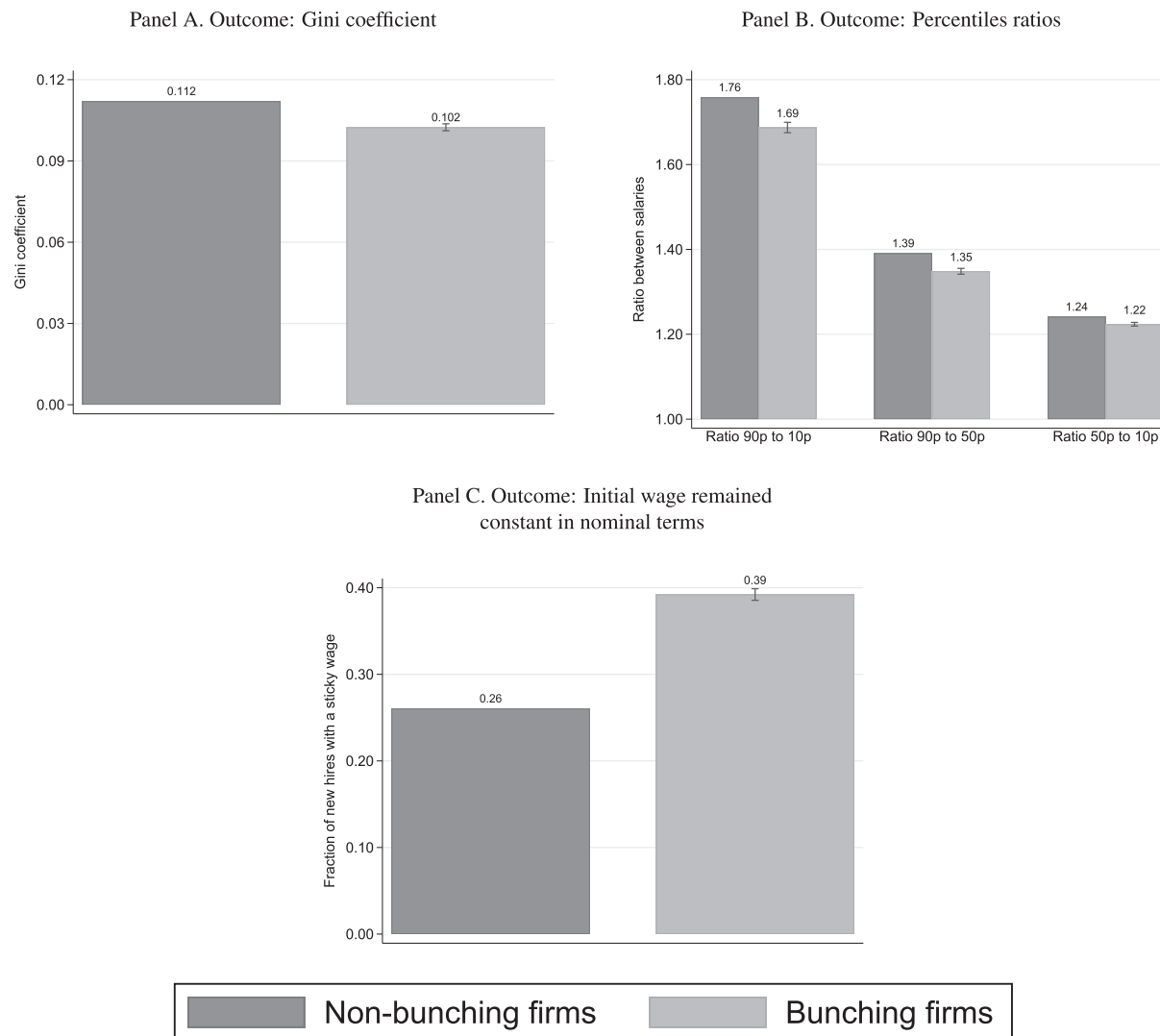
Figure 3 shows that bunching firms tend to compress wage differentials among new hires (see appendix table A9 for the corresponding regression coefficients). The average Gini coefficient among nonbunching firms is 0.112 (panel A).²⁰

¹⁸Ex ante, the direction is ambiguous. To see this, consider a firm that pays workers their fully optimal salary rounded to the nearest 1,000 deciding the wages of two new hires. If the workers' fully optimal salaries are R\$700 and R\$1,400, but the firm pays both of them R\$1,000, then the coarse pricing generates wage compression. Instead, if the first worker's fully optimal salary is R\$1,700, the paid salaries would be R\$2,000 and R\$1,000, respectively. In this case the coarse wage setting increases wage dispersion.

¹⁹The Gini measures overall inequality in the contracted salary distribution, whereas the ratios measure inequality at different parts of the distribution (e.g., top- or low-end inequality; see Lemieux, 2008).

²⁰By country standards, this is a very low level of inequality. The most egalitarian countries in the world—typically, the Nordic countries—have a Gini coefficient on the order of 0.25. Two reasons explain the difference in magnitudes. First, country-level inequality is typically measured using household consumption per capita as the welfare measure, whereas I compute the Gini using earnings. Second, I calculate the Gini among new hires

FIGURE 3.—WAGE COMPRESSION AND WAGE STICKINESS IN THE SALARIES OF NEW HIRES



Dark bars plot the average of the variable listed in the panel title for nonbunching firms. Light bars plot the sum of this average and the estimated $\hat{\beta}$ from equation (1). To calculate the effect of bunching firms on each outcome, I estimate equation (1) at the firm level using as the dependent variable one of the four measures of inequality or the measure of wage stickiness. The regressions control for firm age, share of employees with completed high school and college, education of the firm manager, a dummy for having an HR department, the mean earnings of the firm employees, and fixed effects for firm size, number of hires, and industry-by-microregion. The wage inequality regressions are estimated at the firm level for firms that hired at least two workers in the sample. The wage rigidity regressions are estimated at the worker-by-firm-by-year level and additionally control for worker gender, race, and occupation. The vertical lines denote the 95% confidence interval on the bunching firm dummy using heteroskedasticity-robust standard errors clustered at the firm level.

Bunching firms have a 0.01 lower Gini coefficient (or 8.9% of the baseline value). The decline in overall wage inequality is driven mostly by top- and mid-end inequality (panel B). The ratio between the 90th and 10th percentile is, on average, 3.9% lower (from a baseline ratio of 1.76) in bunching firms relative to the rest of the firms. Similarly, bunching firms have, on average, a 2.8% lower 90th to 50th percentile ratio and a 1.6% lower 50th to 10th percentile ratio than non-bunching firms (from baseline ratios of 1.39 and 1.24, respectively). These effects are robust to excluding small firms (appendix figure A5).

of a given firm, which is likely a more homogeneous population than the overall population of a country.

B. Nominal Wage Rigidity

Nominal wage stickiness influences the effects of monetary policy (Barattieri et al., 2014). Previous work has documented that behavioral considerations such as inertia (Eichenbaum et al., 2011), managerial inattention (Ellison et al., 2018), and fairness norms (Kaur, 2019) influence nominal rigidities. Coarse wage setting might contribute to wage rigidity if it makes firms less likely to change the initial salary of their new hires. To assess this, I estimate equation (1) using as the dependent variable a dummy that equals one if the nominal salary of a new hire remained constant in nominal terms during the year following the hiring, and zero otherwise.

The initial salaries of bunching firms' workers tend to be stickier (figure 3, panel C). From a baseline of 26.0%, workers employed by bunching firms have a 13-percentage-point increase in the probability of experiencing no salary change. Thus, relative to new hires of nonbunching firms, those employed by bunching firms are about 50% more likely to exhibit nominal wage stickiness. This effect is robust to excluding small firms (appendix figure A5).

C. Minimum Wage Spillovers

Dube et al. (2020) hypothesize that in the presence of firms that pay round-numbered wages, a change in the minimum wage could generate a novel spillover effect if the new minimum wage crosses a round number. Intuitively, a change in the minimum wage might cause firms that initially pay a round-numbered wage to fully optimize. However, their data do not allow them to test this hypothesis. In my sample, I observe hiring decisions under fifteen different federal minimum salaries, seven of which are round numbers. I also observe the year $t + 1$ salary of workers hired in year t , which allows me to assess the importance of this potential spillover effect.

I describe the methodology and results in detail in appendix G. In short, using a differences-in-differences approach comparing salaries directly affected by the change in the minimum salary and those not directly affected by it, I find that an increase in the minimum salary reduces the share of round-numbered salaries by 5.4 percentage points (or 11.3%). This finding suggests that changes in the minimum wage can have sizable spillover effects on firm wage-optimization behavior.

VII. Discussion

Setting the right wage is challenging. To estimate the fully optimal wage prescribed by economic models, a firm needs substantial information, including an estimate of the worker's contribution to the firm. Most workers have multiple goals and no measured output, which makes productivity difficult to estimate. This paper posits that the stark bunching at round numbers in the earnings distribution partly reflects the challenges associated with optimal labor pricing. In the data, millions of workers are hired at round-numbered salaries, reflecting a behavior that cannot be accommodated by existing wage-setting models. The evidence presented in this paper indicates that this behavior is partly due to firms engaging in coarse wage setting.

An important unresolved question is whether the coarse wage setting is suboptimal. Establishing optimal pay-setting practices likely requires substantial resources. If these costs are large, offering a coarse wage might lead to better outcomes. Nonetheless, the findings have intrinsic value for understanding how firms set wages. Coarse wage setting may also have consequences for wage inequality and nominal

wage rigidity, and may interact with policies that affect the wage distribution.

Future work could also explore the extent to which rounding reflects the quality of management practices. Management quality is often not available in traditional data sets (the World Management Survey is a notable exception; see Bloom & Van Reenen, 2007). If coarse pricing partly reflects how human resources are managed at the firm, researchers could use the type of salaries offered to new hires as a proxy for overall HR management quality.

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